

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PNG 410 — Past Questions & Materials
by Renance

Course: PNG 410
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 400

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Paper 1 of 1 — PNG 410 - Applied Reservoir Engineering

Level 400 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES QUESTION 2

COLLEGE OF TECHNOLOGY |

DEPARTMENT OF PETROLEUM ENGINEERING The result of a gas chromatographic test performed at FUPRE vabrtist

FIRST SEMESTER EXAMINATION, 2024-2025 SESSION on a gas sample collected from a gas field in the Niger Delta region is

cecerpavetscsqsenenaeneciSbeeneuseeSONGrBEnentnanicenuseuans shown in the table below

pussecemannnsecncsunnouaner

Date: 13-04-2025 TIME ALLOWED: 3HR UNIT: 3 Components Mole Percent (xi)

Nitrogen 0.18

INSTRUCTION: ANSWER THREE QUESTION IN SESSION A AND Carbon dioxide 0.91

TWO IN SESSION B Methane 87.43

Ethane 4.44

SESSION A Propane 3.65

QUESTION 1 i-butane 0.64

. . ; n-butane 1.50

An active water drive reservoir has a cumulative water production of 28640 i-pentane 0.45

bbl was estimated after the first year of production. The pressure profile, Berita 0.38

reservoir and aquifer properties are given as: 4 = ae 0.30

Time (yrs) 0 7 5) 3 4 5 Heptane Plus 0.15

Pressure (psia) | 5027 | 5016 5014 4966 4818 4775 Additional information:

Wen 5.3168 | 8.7385 | 11.7766 | 14.6216 | 17.3312 -

Initial res 1 ; j

Aquifer permeability, K,, 160 mD ervolr alee Pi sina

Water viscosity, 1,, 0.78 cp Reservoir thickness, h ! 55 ft

Drainage radius, r, 9500 ft Connate water saturation, Swe 23%

Aquifer radius, 1, 47500 ft Area, A 410 acres

Total compressibility, C, 3.80×10^{-5} psi⁻¹ Pores

Porosity, 0 220% On osity, @ 19.8%

Aquifer angle 120° Keservoir temperature 200°F

Calculate cumulative water influx for each year |. Calculate the cumulative gas produced at 2500 psia

i. Calculate the recovery factor at 250Upsia

4 Ultimate oil recovery (Nu) to economic limit Gos = $827e^{-0.1626}42$ — 417.52 sth/month

4} Production rate (qos) at end of year 2008 Hint:

2.309kt 1.119/@h 2

Solution Ly Dy Cow To? 119f OAC, Te

The effective rate is: $6uC.r2 @uC,r2$

— p, +706 HE, SHE T _

4(0.000264)kt kh 0.001056kt

- 73 — 827

= Wa Miss _ 973 B27 _ 9 1501 /yr = 15.01%/IT

D, 'h

Nominal decline rate Is:

$e^{7\pi} = 1 - D, 'h$ n

$D, = -\ln(1 - D, 'h) = \ln(1 - 0.1501) = 0.1626/\text{yr} = 16.26\%/\text{yr AV}, == (\text{An t+ Anta}) =) \text{ Mi}$

2 t=1

The remaining reserve:

n

h

$N_{qi} - q_a Ch - \det(973 - 190) J_{el} * 12\text{months}/, AV) = 3 (\text{An} + \text{Anti} + J\text{An pects.}) = >. \text{Avex}$

a a a 0.1626/yr "

= 57785.97 stb h i a 7 aq fs _ (%2)""

$V_i, 5a(2-) \sim (1 - b)D;$

Abandonment time (time to economic limit) $(1 + 0.5D, \phi) D_d, \text{Maa}, (1 - 2) a$

$1(4) = t_n i^* \text{rind: } 2 \ln(a + D_2)$

— = = ' '

$t_a = 5 \text{ ""} \backslash \text{Gn}, 0.1626 \text{ " ra) aemial ie ,}$

The ultimate oil recovery to economic limit is

$N_e, = N_{ag} + N_p = 80519 + 57785.97 = 138304.97 \text{ seb}$

The production rate at the end of 2008 is

_ —D, t

os = dose ~*

do4 = 827 stb/month

a

vA

yr

QUESTION 3

y State four factors affecting the productivity index

1) You are provided with the following data: 2 The figure below shows an active and an observation well,

Calculate the pressure change in the observation well at:

(Area of the field | 5400 acres |

; he active well after 10 hrs of producti

Thickness _ adistance of from t U production

1 i oune eet | 37 ft | ' The wellbore after 10 hrs of production

orosity of the formation 20.1% for the top 12ft , = -

| 19.4% for the middle 17ft BA :
 | 17.6% for the bottom 8ft Observation —
 | Water saturation | 26.3% for the top 12ft well 420A any
 | ie for the middle 17ft
 \ aaa 18.9% for the bottom 8ft
 \ | Gas oil ratio {500 cuft/bbl | Wellbore radius, %, 0.48 ft
 | Net reservoir sand thickness | 37ft | Total compressibility, C_e 18.9×10^{-10} psi'
 | Gravity [0.68 _ Initial pressure, P , 3860 psi
 Gravity f 35°AP | Formation permeability, k 997 mD
 = . q ft Oil viscosity, μ_o 0.86 cp
 | thi er eer ve San | 40 | Oil formation volume factor, B_o 1.132 rb/stb
 ICKNESS | Porosity, ϕ 20.7%
 y os 1175
 By = $0.972 + 0.000147 \ln(2) + 12st$] SESSION B
 Yo QUESTION 5

- Calculate the oil initially in place
- Calculate the stock tank oil initially in place The result Udua gas field volumetric analysis and production data is
- Calculate the STOIP if a cut-off value of 10% is applied to given below i
 the porosity and 15% to the water saturation. Parameter | Value a
 ultimate recoverable gas reserves | 37.82
 | (MMscf)
 restricted production rate (MMscf/month) 1385
 (ii) Use one diagram to show the variation of R_s , Y_o , B_y and B_o nominal decline rate (per month)
 10.035 |
 Economic limit (MMscf/month) 23
 with pressure
 Collis the following, assuming exponential decline
 \. The life of the well,

QUESTION 4 i. The prorated time

iii. Cumulative gas production during the restricted rate Initial reservoir pressure = 2740 psi
 Current reservoir pressure = 2460 psi

; declining. Initial oil FVF = 4.3985 rb/stb Initial gas exp. Factor = 198 3 scffcuft

iv. The cumulative production the year production starts Initial solution GOR = 643 ect/stb +5 ala
 ee 14 enteun

oil FVF = 1.3578 rbistb Cum. Gas Produced = 15498 MMscf

QUESTION 6 Current solution GOR = 577.3 scf/stb Gas/oil sand volume ratio = 0.7

Cum. Oil Produced = 18. 9 MMstb Cum. Water produced = 3.3 MMstb

(a) Calculate the average reservoir pressure at the Datum depth of Connate water saturation =
 17% Formation compressibility = 3.5×10^{-10} psi!

8750 ftss for the following fluid pressure gradients, given that the STOIIP = 120 MMstb Water compressibility = 3.5×10^{-6} pst!

GOC and OWC are at 8700 ftss and 8800 ftss respectively: ; /

The production trend of Okoso oil field is presented in the figure

P ip dP ; Por ; .

(S) = 0.08 psi/ft, (=) = 0.269 psi/ft, (5) below. The estimated economic limit of this field is 190 stb/month.

gas ott water

= 0.434 psi/fet 10,000

_ 973 STB/month

1A= t 8690 ft .

Well A = 3685 psig a a - = (1 January 2003) 827 STB/month

Well C = 3725 psig at 8820 ftss z Mi AWA ho

Well D = 3689 psig at 8710 ftss = 1,000; -*.-*.-. Malina foi mem eens j

Well E = 3713 psig at 8790 ftss i i) a ae

(b) A well penetrates a reservoir near the top of a fluid column. The 8

GOC has been detected by logs but the OWC was not. An oll 3 yoo} ----- Cumulative oil production (W,) _ |

sample was taken at 7890 ft TVD with a pore pressure of 3890 psig 3 31 December 01 =80,519 STB recorded. The field water gradient is 0.445 psi/ft, oil gradient is a

0.315 psifft. find the OWC 5

10 =

QUESTION 7 o7 an Fa ne

Year

Analyze the C4 reservoir with production and fluid data given j

n

below Based on your findings, recommend the C4 reservoir toe table Calculate -

which secondary recovery technique to adopt to further recover the Pei 2, Effective (D') and nominal (D) decline rates

il

4, Remaining reserves (Np) from 1 January 2003 to the field's

economic limit

, Time (f) to produce to economic limit